The first asymptotic results for OEIS A112509

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Abstract

For a binary string s of length n, let $f(s)$ be the number of distinct non negative integers represented by contiguous substrings of s, interpreted in binary. Let

$$a(n) = \max_{|s|=n} f(s).$$

Prior to the present work, no asymptotic results appear to have been available for the sequence $a(n)$, although exact values had been computed for small n. In this paper we prove that

$$a(n) \geq \frac{1}{2}n^2 - O(n^{3/2}),$$

and hence $a(n)/n^2 \to 1/2$. The proof is completely explicit. We also prove that there exist near optimal strings with density of ones equal to $1 - O(n^{-1/2})$, and that every optimal string has density of ones equal to $1 - O(n^{-1/4})$. In addition, we derive upper bounds which penalise both the number and the run structure of the zeros.

1 Introduction

For a binary string $s \in \{0,1\}^n$, let $f(s)$ denote the number of distinct non negative integers represented by its contiguous substrings. Define

$$a(n) = \max_{|s|=n} f(s).$$

This is OEIS sequence A112509.

The most obvious upper bound is obtained by counting substrings:

$$a(n) \leq \frac{n(n+1)}{2}.$$

Thus any quadratic asymptotic for $a(n)$ can have leading constant at most 1/2. The main result of the paper shows that this upper bound is asymptotically sharp.

Theorem 1. *As $n \to \infty$,*

$$a(n) \geq \frac{1}{2}n^2 - O(n^{3/2}).$$

Consequently,

$$\lim_{n\to\infty} \frac{a(n)}{n^2} = \frac{1}{2}.$$

We shall actually prove an explicit quantitative version with a concrete constant.

2 Definitions and elementary bounds

If $w = w_1 \cdots w_k \in \{0,1\}^k$, define its binary value by

$$\nu(w) = \sum_{i=1}^{k} w_i 2^{k-i}.$$

Leading zeros do not affect the value, so for example $\nu(01) = \nu(1) = 1$.

If $s = s_1 \cdots s_n$, a substring of s is a word of the form $s_i s_{i+1} \cdots s_j$ with $1 \le i \le j \le n$. Let $V(s)$ be the set of all values represented by substrings of s:

$$V(s) = \{\nu(s_i \cdots s_j) : 1 \le i \le j \le n\}.$$

Then

$$f(s) = |V(s)|.$$

Lemma 2. *For every $n \ge 1$,*

$$a(n) \le \frac{n(n+1)}{2}.$$

In particular,

$$\limsup_{n\to\infty} \frac{a(n)}{n^2} \le \frac{1}{2}.$$

Proof. A distinct positive represented integer has a unique standard binary expansion beginning with 1. Therefore each distinct positive represented integer is represented by at least one substring beginning with 1, and different positive integers correspond to different such substrings. Hence the number of distinct positive represented integers is at most the number of substrings beginning with 1, which is at most the total number of substrings, namely $n(n+1)/2$.

If $0 \in V(s)$, this contributes at most one additional value, but then at least one substring begins with 0, so the number of substrings beginning with 1 is at most $n(n+1)/2 - 1$. Thus even in this case the total number of represented values is at most

$$\left(\frac{n(n+1)}{2} - 1\right) + 1 = \frac{n(n+1)}{2}.$$

If $0 \notin V(s)$, then there is no extra value to add. Therefore $f(s) \le n(n+1)/2$ for every s, and taking the maximum gives the result. □

3 An explicit construction

We now construct binary strings with very large distinct value sets.

Lemma 3. *Let $k \ge 2$ and $M \ge k(k+1)/2$. Then there exist strictly decreasing positive integers*

$$b_1 > b_2 > \cdots > b_k \ge 1$$

with

$$\sum_{i=1}^{k} b_i = M.$$

Moreover, they may be chosen so that all b_i differ from the average M/k by at most k:

$$\left| b_i - \frac{M}{k} \right| \le k \qquad (1 \le i \le k).$$

Proof. Set

$$q = \left\lfloor \frac{M}{k} - \frac{k-1}{2} \right\rfloor$$

and define the remainder

$$r = M - \left(kq + \frac{k(k-1)}{2} \right).$$

Since the fractional part of $M/k - (k-1)/2$ lies in $[0,1)$, we have $0 \le r < k$.

Now define

$$b_i = \begin{cases} q + k - i + 1, & 1 \le i \le r, \\ q + k - i, & r < i \le k. \end{cases}$$

Then $b_1 > b_2 > \cdots > b_k$, and a direct summation gives

$$\sum_{i=1}^{k} b_i = \left(kq + \frac{k(k-1)}{2} \right) + r = M.$$

Also, since $M \ge k(k+1)/2$, we have

$$\frac{M}{k} - \frac{k-1}{2} \ge 1,$$

so $q \ge 1$ and hence $b_k \ge q \ge 1$.

Finally, the sequence (b_i) lies in an interval of length less than k, namely between q and $q+k$. Since its average is M/k, every term lies within distance at most k from the average. This proves the final claim. □

Fix $n \ge 64$ and define

$$k = \left\lfloor \frac{\sqrt{n}}{4} \right\rfloor, \qquad M = n - (k+1).$$

By $n \ge 64$ we have $\sqrt{n}/4 \ge 2$, hence $k \ge 2$. Also,

$$M = n - (k+1) \ge n - \frac{\sqrt{n}}{4} - 1 \ge \frac{n}{2}$$

for $n \ge 64$, and therefore $M \ge k(k+1)/2$. So Lemma **??** applies.

Choose strictly decreasing positive integers $b_1 > b_2 > \cdots > b_k$ summing to M, and define

$$s = 1^{b_1} 0\, 1^{b_2} 00\, 1^{b_3} 0 \cdots 0\, 1^{b_k}. \tag{1}$$

Then $|s| = n$.

4 The injective family

For integers i, j, α, β satisfying

$$1 \le i < j - 1 \le k - 1, \qquad 1 \le \alpha \le b_i, \qquad 1 \le \beta \le b_j,$$

let $u_{i,j,\alpha,\beta}$ be the substring of s which starts in the ith block of ones, with exactly α ones remaining in that block, and ends in the jth block of ones, after exactly β ones of that block. Since $j \ge i+2$, this substring contains at least one full interior one-block.

Lemma 4. *The map*

$$(i, j, \alpha, \beta) \longmapsto u_{i,j,\alpha,\beta}$$

is injective.

Proof. Suppose

$$u_{i,j,\alpha,\beta} = u_{i',j',\alpha',\beta'}.$$

We compare the run lengths of the blocks of ones occurring in this common word.

Because $u_{i,j,\alpha,\beta}$ begins inside the ith one-block and ends inside the jth one-block, the full interior one-blocks occurring in it are exactly

$$b_{i+1}, b_{i+2}, \ldots, b_{j-1}.$$

Likewise, the full interior one-blocks occurring in $u_{i',j',\alpha',\beta'}$ are exactly

$$b_{i'+1}, b_{i'+2}, \ldots, b_{j'-1}.$$

Since the two substrings are equal as binary words, these consecutive interior block sequences must be equal:

$$b_{i+1}, b_{i+2}, \ldots, b_{j-1} = b_{i'+1}, b_{i'+2}, \ldots, b_{j'-1}.$$

But the global sequence $b_1 > b_2 > \cdots > b_k$ is strictly decreasing, so any consecutive subsequence occurs in exactly one place. Therefore

$$i = i' \qquad \text{and} \qquad j = j'.$$

Once i and j are determined, the initial run length of ones in the substring is exactly α, and the final run length of ones is exactly β. Equality of the substrings therefore implies

$$\alpha = \alpha', \qquad \beta = \beta'.$$

Thus the map is injective. □

5 Quantitative lower bound

Theorem 5. *For every $n \geq 64$,*

$$a(n) \geq \frac{1}{2}n^2 - 15n^{3/2}.$$

Consequently,

$$a(n) \geq \frac{1}{2}n^2 - O(n^{3/2}).$$

Proof. Let s be the string constructed in (**??**). By Lemma **??**, all substrings $u_{i,j,\alpha,\beta}$ are distinct. For each pair (i, j) with $1 \leq i < j - 1 \leq k$, there are exactly $b_i b_j$ choices of (α, β). Therefore

$$f(s) \geq \sum_{1 \leq i < j-1 \leq k} b_i b_j.$$

Now

$$\sum_{1 \leq i < j-1 \leq k} b_i b_j = \sum_{1 \leq i < j \leq k} b_i b_j - \sum_{i=1}^{k-1} b_i b_{i+1}.$$

Also,

$$\sum_{1\le i<j\le k} b_i b_j = \frac{1}{2}\left(\left(\sum_{i=1}^{k} b_i\right)^2 - \sum_{i=1}^{k} b_i^2\right).$$

Since $\sum b_i = M = n-(k+1)$, we obtain

$$f(s) \ge \frac{1}{2}(n-k-1)^2 - \frac{1}{2}\sum_{i=1}^{k} b_i^2 - \sum_{i=1}^{k-1} b_i b_{i+1}. \tag{2}$$

We now estimate the three error terms.

First, since $k = \lfloor \sqrt{n}/4 \rfloor$, we have

$$k \le \frac{\sqrt{n}}{4}.$$

Hence

$$\frac{1}{2}(n-k-1)^2 \ge \frac{1}{2}n^2 - n(k+1) \ge \frac{1}{2}n^2 - \frac{1}{4}n^{3/2} - n.$$

Because $n \ge 64$, we have $n \le n^{3/2}/8$. Therefore

$$\frac{1}{2}(n-k-1)^2 \ge \frac{1}{2}n^2 - \frac{3}{8}n^{3/2}. \tag{3}$$

Next, because $k = \lfloor \sqrt{n}/4 \rfloor$ and $\sqrt{n}/4 \ge 2$, we also have

$$k \ge \frac{\sqrt{n}}{8}.$$

Indeed, if $x \ge 2$ then $\lfloor x \rfloor \ge x/2$.

By Lemma **??**, each b_i differs from the average M/k by at most k. Hence

$$\left| b_i - \frac{M}{k} \right| \le k.$$

Writing $b_i = M/k + d_i$ with $|d_i| \le k$, we get

$$\sum_{i=1}^{k} b_i^2 = \sum_{i=1}^{k}\left(\frac{M}{k} + d_i\right)^2 = \frac{M^2}{k} + \sum_{i=1}^{k} d_i^2 \le \frac{M^2}{k} + k^3.$$

Since $M \le n$ and $k \ge \sqrt{n}/8$, this gives

$$\frac{M^2}{k} \le 8n^{3/2}.$$

Also, since $k \le \sqrt{n}/4$,

$$k^3 \le \frac{1}{64}n^{3/2}.$$

Therefore

$$\sum_{i=1}^{k} b_i^2 \le \frac{513}{64}n^{3/2} < 9n^{3/2}. \tag{4}$$

Finally, for the adjacent product term we use the elementary inequality $xy \le (x^2+y^2)/2$:

$$\sum_{i=1}^{k-1} b_i b_{i+1} \le \frac{1}{2}\sum_{i=1}^{k-1}(b_i^2 + b_{i+1}^2) \le \sum_{i=1}^{k} b_i^2.$$

Combining this with (**??**), we obtain

$$\sum_{i=1}^{k-1} b_i b_{i+1} < 9n^{3/2}. \tag{5}$$

Substituting (**??**), (**??**), and (**??**) into (**??**) yields

$$f(s) \geq \left(\frac{1}{2}n^2 - \frac{3}{8}n^{3/2}\right) - \frac{1}{2} \cdot 9n^{3/2} - 9n^{3/2} > \frac{1}{2}n^2 - 14n^{3/2}.$$

In particular,

$$f(s) \geq \frac{1}{2}n^2 - 15n^{3/2}.$$

Since $a(n) \geq f(s)$, the theorem follows. □

Corollary 6. *The limit*

$$\lim_{n\to\infty} \frac{a(n)}{n^2}$$

exists and is equal to $1/2$.

Proof. By Lemma **??**,

$$\frac{a(n)}{n^2} \leq \frac{n+1}{2n} \to \frac{1}{2}.$$

By Theorem **??**,

$$\frac{a(n)}{n^2} \geq \frac{1}{2} - \frac{15}{\sqrt{n}} \to \frac{1}{2}.$$

The squeeze theorem gives the result. □

6 Density of ones in near optimal strings

The explicit construction immediately yields near optimal strings with density of ones tending to 1.

Proposition 7. *There exists a family of binary strings* s_n *such that*

$$f(s_n) \geq \frac{1}{2}n^2 - O(n^{3/2})$$

and

$$\frac{N_1(s_n)}{n} = 1 - O(n^{-1/2}).$$

Equivalently,

$$N_0(s_n) = O(\sqrt{n}).$$

Proof. Take s_n to be the explicit string from (**??**). By Theorem **??**,

$$f(s_n) \geq \frac{1}{2}n^2 - O(n^{3/2}).$$

The string contains exactly $k+1$ zeroes: one in every separator except the second, which contributes one extra zero. Since

$$k = \left\lfloor \frac{\sqrt{n}}{4} \right\rfloor,$$

we have

$$N_0(s_n) = k + 1 = O(\sqrt{n}).$$

Therefore

$$N_1(s_n) = n - N_0(s_n) = n - O(\sqrt{n}),$$

and dividing by n gives

$$\frac{N_1(s_n)}{n} = 1 - O(n^{-1/2}).$$

□

7 Upper bounds in terms of zeros

We now derive general upper bounds for an arbitrary binary string in terms of the number, positions, and run structure of its zeroes.

Lemma 8. *Let $s \in \{0,1\}^n$, and let its zero positions be*

$$p_1 < p_2 < \cdots < p_z.$$

Then

$$f(s) \leq \frac{n(n+1)}{2} - \sum_{r=1}^{z}(n - p_r + 1).$$

In particular,

$$f(s) \leq \frac{n(n+1)}{2} - \frac{z(z+1)}{2}.$$

Proof. For each $x \in V(s)$, choose one substring w_x of s with $\nu(w_x) = x$, subject to the condition that w_x has minimal length among all substrings representing x.

The map

$$x \longmapsto w_x$$

is injective. Indeed, if $w_x = w_y$ as binary strings, then

$$x = \nu(w_x) = \nu(w_y) = y.$$

Now suppose $x > 0$. Then the first bit of w_x must be 1. For if the first bit were 0, then deleting all leading zeros from w_x would produce a strictly shorter substring of s with the same value, contradicting the minimality of w_x.

It follows that every positive represented integer has a chosen representative beginning with 1. Hence the number of distinct positive represented integers is at most the number of substrings of s beginning with 1.

There are exactly $n(n+1)/2$ substrings in total. For each zero position p_r, exactly $n - p_r + 1$ substrings begin at that position, and none of them begins with 1. Therefore the number of substrings beginning with 1 is at most

$$\frac{n(n+1)}{2} - \sum_{r=1}^{z}(n - p_r + 1).$$

If $0 \notin V(s)$, this already bounds $f(s)$. If $0 \in V(s)$, then the one-bit substring 0 occurs at some zero position and is among the substrings excluded in the count above. Thus including the value 0 does not increase the right hand side. In either case,

$$f(s) \leq \frac{n(n+1)}{2} - \sum_{r=1}^{z}(n - p_r + 1).$$

For fixed z, the right hand side is maximised when the zeroes are as far to the right as possible, namely when

$$p_r = n - z + r \qquad (1 \leq r \leq z).$$

Then

$$\sum_{r=1}^{z}(n - p_r + 1) = 1 + 2 + \cdots + z = \frac{z(z+1)}{2},$$

which yields the second inequality. □

Corollary 9. *If s_n is an optimal string of length n and $z_n = N_0(s_n)$, then*

$$z_n = O(n^{3/4}).$$

Equivalently,

$$\frac{N_1(s_n)}{n} = 1 - O(n^{-1/4}).$$

Proof. Let s_n be optimal. Then by Theorem **??**,

$$\frac{1}{2}n^2 - 15n^{3/2} \leq a(n) = f(s_n).$$

By Lemma **??**,

$$f(s_n) \leq \frac{n(n+1)}{2} - \frac{z_n(z_n+1)}{2}.$$

Combining the two inequalities gives

$$\frac{z_n(z_n+1)}{2} \leq 15n^{3/2} + \frac{n}{2} + 1.$$

Thus $z_n^2 = O(n^{3/2})$, so $z_n = O(n^{3/4})$. Since $N_1(s_n) = n - z_n$, dividing by n yields the density statement. □

8 A run sensitive upper bound

Let the maximal zero-runs of a binary string s have lengths $c_1, \ldots, c_m$, and let the ith zero-run begin at position a_i. Define r_i to be the number of positions strictly to the right of the ith zero-run. Equivalently,

$$r_i = n - (a_i + c_i) + 1.$$

Lemma 10. *With the above notation,*

$$f(s) \leq \frac{n(n+1)}{2} + 1 - \sum_{i=1}^{m}\left(c_i(n - a_i - c_i + 1) + \binom{c_i+1}{2}\right).$$

Equivalently,

$$f(s) \leq \frac{n(n+1)}{2} + 1 - \sum_{i=1}^{m}\left(c_i r_i + \binom{c_i+1}{2}\right).$$

Proof. Apply Lemma **??** and group the sum by zero-runs. If the ith run occupies positions

$$a_i, a_i + 1, \ldots, a_i + c_i - 1,$$

then its contribution to the sum in Lemma **??** is

$$\sum_{t=0}^{c_i-1} (n - a_i - t + 1) = c_i(n - a_i + 1) - \frac{c_i(c_i - 1)}{2}.$$

Now

$$r_i = n - (a_i + c_i) + 1 = n - a_i - c_i + 1,$$

so

$$c_i(n - a_i + 1) - \frac{c_i(c_i - 1)}{2} = c_i r_i + \binom{c_i + 1}{2}.$$

Summing over all zero-runs gives the result. □

Corollary 11. *If a binary string s has z zeroes arranged in m maximal zero-runs, then*

$$f(s) \leq \frac{n(n+1)}{2} + 1 - \frac{z(z+1)}{2} - \frac{m(m-1)}{2}.$$

Proof. By Lemma **??**,

$$f(s) \leq \frac{n(n+1)}{2} + 1 - \sum_{i=1}^{m} \left(c_i r_i + \binom{c_i + 1}{2} \right).$$

Since distinct zero-runs are separated by at least one 1, for $i < m$ we have

$$r_i \geq \sum_{j>i} c_j + (m - i),$$

while $r_m \geq 0$. Therefore

$$\sum_{i=1}^{m} c_i r_i \geq \sum_{i<j} c_i c_j + \sum_{i=1}^{m-1} c_i(m - i).$$

Hence

$$\sum_{i=1}^{m} \left(c_i r_i + \binom{c_i + 1}{2} \right) \geq \sum_{i<j} c_i c_j + \sum_{i=1}^{m} \binom{c_i + 1}{2} + \sum_{i=1}^{m-1} c_i(m - i).$$

Now

$$\sum_{i<j} c_i c_j + \sum_{i=1}^{m} \binom{c_i + 1}{2} = \binom{z + 1}{2},$$

because $z = \sum c_i$ and

$$\binom{z + 1}{2} = \frac{z^2 + z}{2} = \sum_{i<j} c_i c_j + \sum_{i=1}^{m} \frac{c_i^2 + c_i}{2}.$$

It remains to bound the final term. Since each $c_i \geq 1$, the quantity

$$\sum_{i=1}^{m-1} c_i(m - i)$$

is minimised when $c_1 = \cdots = c_{m-1} = 1$ and all remaining zeroes are placed in the final run. In that case the minimum is

$$(m-1) + (m-2) + \cdots + 1 = \binom{m}{2}.$$

Therefore

$$\sum_{i=1}^{m} \left(c_i r_i + \binom{c_i + 1}{2} \right) \geq \binom{z+1}{2} + \binom{m}{2},$$

and the claimed bound follows. □

9 Computational evidence

The theoretical results above are fully rigorous and self contained. We briefly record the main computational findings which motivated them.

Exact values of $a(n)$ were computed for $1 \leq n \leq 150$, extending the previously known range, and lower bounds were obtained up to $n = 2 \times 10^9$. In particular, the ratio $a(n)/n^2$ rises from 0.3596 at $n = 50$ to 0.4031 at $n = 150$, while the large scale lower bounds already give $a(n)/n^2 \geq 0.49996$ at $n = 2 \times 10^9$. Three independent computational methods were compared on overlapping ranges: unconstrained Metropolis–Hastings search, constrained Metropolis–Hastings search, and exact exhaustive search over a run length template. On the overlap where more than one source was available, the best values agreed across methods. The unconstrained Metropolis–Hastings search was particularly effective: in the range $n \leq 150$, at least one out of ten independent unconstrained restarts reached the optimum for every n considered.

The updated extended results also show that the set of optimal strings has very low empirical entropy. If S_n denotes the set of optimal strings and $H(n) = \log_2 |S_n|$, then for $n \leq 150$ the quantity $H(n)$ is almost always zero or very small, remains below 4, and the optimal strings typically share long common prefixes. Likewise, for $n \leq 150$ the longest common zero-separator prefix is completely rigid in 28 cases, varies by at most one separator in 119 further cases, and shows larger variation only at the three values $n = 51, 70, 116$.

10 Heuristic remarks

The proofs in this paper use only the explicit construction and the zero-penalty arguments above. Nevertheless, the computational data suggests a clear heuristic mechanism. The objective is to maximise the number of substrings beginning with 1 that remain distinct after leading zeros are stripped. Long repeated patterns create large common-prefix collisions, so strings with many short repeated blocks perform poorly. By contrast, the explicit construction front loads the ones into a small number of long decreasing blocks, separated by very short zero-runs. This creates many substrings while suppressing long scale repetition. The empirically observed optimal strings appear to exploit the same principle, but in a more refined form.